Crossing Is Not Insertion: Recoverability-Aware Evaluation in 2D Orbital Control

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Abstract

This study asks a focused question. Can target-radius crossing alone indicate simulator-defined recoverability in simplified two-dimensional orbital control under the tested benchmark conditions? A trajectory may touch the target radius yet retain radial or tangential velocity errors, much like a runner reaching a line without settling into a stable pace. The evidence comes from controlled simulation. A 24-case benchmark compared explicit controllers, post-cross synchronization, transfer variants, diagnostic modifications, PPO, and imitation-learning baselines. The Phase31-style reference produced 8/24 crossings but no recoverable crossings, whereas Phase34 preserved 8/24 crossings and increased recoverable crossings to 8/24 through post-cross regulation of the existing crossing-producing cases. Phase36B showed no basin expansion. All four families returned 8/24 crossings and 8/24 recoverable crossings, with no overspeed or instability flags. It is worth noting that identical counts do not establish identical trajectories, broader robustness, or formal equivalence across controllers beyond this designed benchmark grid and simulator setting. The Phase37 diagnostics remained more limited. Their method-specific outcomes, along with the learning experiments, should not be treated as universal evidence against those approaches. In my opinion, the main contribution is evaluative discipline: supported by the available data, crossing alone appears insufficient, while broader robustness and mission relevance remain subject to further research.

1. Introduction and Research Objectives

1.1 Background and Current Research Context

Orbital control is more than geometry. A useful orbital state depends on position and velocity evolving together, rather than on radius alone [1]. In my opinion, a spacecraft that merely crosses the desired radius resembles a car entering the correct lane while still moving too fast to remain there safely.

The simulator deliberately stays simple. It represents planar point-mass gravity, Cartesian state variables, and a bounded two-component thrust command. It is worth noting that this simplicity removes many mission effects, yet it also reveals whether the success metric rewards stable regulation or only brief geometric contact.

Several control traditions address this task. They include explicit guidance, numerical trajectory optimization, PPO-based continuous control, and imitation learning from demonstrations [1]-[4]. However, I think these approaches can appear stronger or weaker depending on the evaluation rule, which raises reasonable doubt about success labels based only on radius crossing.

1.2 Research Gap

The metric can mislead the reader. A radius-only score may reward a trajectory even when its post-cross velocity state remains unfavorable. In my opinion, that is similar to calling a student successful for entering the classroom while ignoring whether the student can follow the lesson afterward.

Two outcomes must be separated. Target-radius crossing records a geometric event, while recoverable crossing additionally requires later entry into a simulator-defined state basin. It is worth noting that this basin constrains radius error, radial velocity, and tangential-velocity error, although the chosen thresholds remain project-specific and perhaps sensitive to future revision.

1.3 Research Questions

The main question is straightforward. **Under simplified 2D orbital dynamics, is target-radius crossing sufficient for simulator-defined recoverable crossing?** In my opinion, this wording keeps the claim testable while avoiding any suggestion that the benchmark represents real mission insertion.

Three related questions guide the experiments. One asks whether post-cross synchronization helps existing crossing cases, while another tests whether upstream variants expand the empirical crossing basin. A third question, perhaps the most uncertain, examines what PPO and imitation learning reveal about transferring long-horizon controller structure under method-specific evaluations.

1.4 Research Objectives

The objectives follow from those questions. The study defines two metrics, builds a controlled benchmark, and compares explicit and learning-based controller families. It is worth noting that the project also records unsuccessful variants, because negative evidence can narrow the design space even when it does not produce a new controller.

1.5 Contribution, Significance, and Scope

The contribution is mainly methodological. The benchmark separates upstream access from downstream state synchronization and links each interpretation to stored evidence. In my opinion, this distinction matters because an attractive trajectory plot can otherwise hide whether a controller created a new crossing or merely improved an already-crossing case.

The scope remains intentionally narrow. The simulator is deterministic, planar, point-mass based, and evaluated on 24 designed initial-condition cases. It is worth noting that the study does not establish formal stability, physical safety, real orbital insertion, or generalization beyond the tested grid, all of which remain subject to further research.

2. Methods

2.1 Overall Research Workflow and Implementation Timeline

The work developed in stages. Simulator construction preceded learning experiments, metric refinement, post-cross control, upstream diagnostics, and final artifact validation. It is worth noting that Phase32 direct shooting and Phase33 structure extraction informed later reasoning, but they were supporting probes rather than primary benchmark results [4].

Table 1. Research Workflow and Implementation Stages

Stage	Main activity	Purpose in this study
Simulator and early control	Built the simplified 2D environment and explicit controllers	Established the experimental platform
Learning experiments	Evaluated PPO, behavior cloning, and imitation-learning variants	Assessed learned-policy transfer
Metric refinement	Separated target-radius crossing from recoverable crossing	Defined the central evaluation problem
Phase31-style reference and Phase34	Compared crossing-only behavior with post-cross synchronization	Established the principal positive result
Phase36	Tested transfer-family variants and diagnosed non-crossing cases	Evaluated empirical crossing-basin expansion
Phase37	Tested radial timing and weak tangential shaping	Produced diagnostic negative evidence

Artifact validation	Consolidated CSV results, figures, and regression checks	Supported traceable reporting

2.2 Simplified 2D Simulation Environment

The environment models planar motion. Its state contains Cartesian position and velocity, while the action is bounded two-dimensional thrust. In my opinion, the model is best viewed as a teaching tool: useful for isolating one mechanism, but too simplified for direct mission claims.

The shared constants remain fixed. They include G = 6.67430e-11, M = 1.989e30, spacecraft mass 722.0, target radius 7.5e12, dt = 100.0, and 100000 maximum steps. It is worth noting that the numerical update advances velocity before position, while omitting attitude dynamics, fuel depletion, uncertainty, delays, and environmental perturbations.

2.3 State, Action, and Derived Quantities

The controller issues bounded thrust commands. Those commands are normalized and converted into acceleration using thrust magnitude and spacecraft mass. I think the derived variables are especially important here, because radius alone cannot reveal whether radial and tangential velocities are approaching a recoverable combination.

Recoverability uses fixed normalized thresholds. The ratios are 2.5e-3 for radius, 2.0e-2 for radial velocity, and 2.5e-1 for tangential-velocity error. It is worth noting that these values support consistent comparison, but the study does not test threshold sensitivity, so their broader physical meaning remains uncertain.

2.4 Data Source, Type, and Scale

All primary data are simulator-generated. The repository stores trajectories, state-action demonstrations, per-rollout rows, aggregate summaries, and figures. In my opinion, this provenance is clear, although it also means the evidence inherits every simplification and bias built into the simulator.

The main evidence uses stored CSV rows. Phase34 and Phase36B each contain 96 rows, Phase37A contains 144, and Phase37B contains 24 subset rows. It is worth noting that learning experiments used separate datasets and protocols, which prevents a clean leaderboard comparison with the unified 24-case benchmark.

2.5 Benchmark Design and Metric Definitions

The benchmark contains 24 cases. It combines three radius ratios, four velocity angles, and two thrust-scale settings. I think this grid is useful for controlled comparison, but its hand-designed structure possibly favors patterns already visible during controller development.

Table 2. Benchmark and Metric Definitions

Item	Definition	Scope / values
Initial radius ratio	Initial radius divided by target radius	0.98, 1.00, 1.02
Initial velocity angle	Initial velocity direction in degrees	150, 165, 170, 175
Thrust scale	Controller thrust-scale setting	8000, 10000
Benchmark size	Cartesian product of the three factors	3 x 4 x 2 = 24 cases
Target-radius crossing	Trajectory reaches or passes through target radius	Counted within stated scope

Recoverable crossing	Crossing followed by later entry into the recoverability basin	Counted within stated scope
Phase34-compatible crossing	Crossing without overspeed or instability under Phase34 handoff assumptions	Diagnostic only
Overspeed	Simulator flag for excessive speed	Safety and regression metric
Instability	Simulator flag for unstable rollout behavior	Safety and regression metric
Closest approach / crossing potential	Proximity diagnostics such as minimum radius error	Not crossing or recoverability

The full benchmark supports several comparisons. Phase34, Phase36B, and Phase37A use all 24 cases, whereas Phase37B uses selected and regression subsets. It is worth noting that subset counts cannot be read as full-benchmark performance, even when the numerator looks similar.

The crossing basin is empirical here. It means the tested initial conditions that produce at least one target-radius crossing. In my opinion, this definition is essential because the word basin might otherwise suggest a formally computed region of attraction or reachability set, which this study does not provide.

2.6 Data Quality Control, Bias, and Reproducibility Boundaries

Several controls improve traceability. Fixed benchmark definitions, shared settings, source-linked plots, stored artifacts, and a regression guard support paper-level checking [5], [6]. It is worth noting that the guard validates recorded outputs rather than rerunning every historical experiment, so complete independent reproducibility has not yet been demonstrated.

Bias remains possible. Development and evaluation occur in the same simulator, the grid is small, and Phase37B uses selected cases. I think these choices make the results useful as controlled evidence, but perhaps not as population-level estimates or proof of robustness across orbital conditions.

2.7 Controller Families

The controllers serve different roles. The Phase31-style reference can produce crossings, while Phase34 adds explicit synchronization after the first crossing. It is worth noting that Phase36B changes upstream transfer behavior, whereas Phase37A and Phase37B probe radial timing and weak tangential shaping without redefining the benchmark.

Learning methods form a separate group. They include behavior cloning, PPO fine-tuning, phase-conditioned imitation, soft phase conditioning, phase stabilization, and minimal imitation learning. In my opinion, their inclusion is informative, although their heterogeneous data and evaluation settings limit any claim that one method outperforms another.

2.8 AI Models and Technical Rationale

PPO fits continuous-control problems. The project used it after behavior-cloning initialization, drawing on a standard policy-gradient approach [2]. However, I think the short fine-tuning setup possibly limited exploration, and the current evidence cannot separate reward design, initialization, and training duration as causes of failure.

Imitation learning uses demonstrations directly. That choice is reasonable when explicit trajectories and phase labels are available [3]. It is worth noting that supervised fit may still accumulate long-horizon error, much like copying each steering correction correctly while gradually drifting off the road.

The models used related outputs. Policies produced two-component thrust actions, while phase-conditioned variants also predicted phase logits or probabilities. In my opinion, these designs are technically relevant, but differing observations and protocols mean the table below should be read as documentation rather than a unified competition.

Table 3. AI and Learning-Based Baselines

Method	Training / data source	Evaluation setting	Verified outcome
Behavior cloning	Balanced simulator-generated explicit-controller dataset	Fixed-baseline transfer comparison	0 crossings; simulator success false
PPO fine-tuning	Short PPO update from behavior-cloning weights	Same fixed-baseline transfer setting	0 crossings; simulator success false
Phase-conditioned imitation	States, actions, and explicit-controller phase labels	Method-specific rollout with phase argmax	0 crossings; simulator success false
Soft phase-conditioned imitation	States, actions, and phase labels	Method-specific rollout with soft phase probabilities	0 crossings; simulator success false
Learned phase stabilization	Loaded learned-phase policy	Stabilization diagnostic; action head unchanged	5 crossings; simulator success false
Minimal imitation learning	Explicit-trajectory state and action arrays	Method-specific minimal-IL rollout	0 crossings; simulator success false

These outcomes remain method-specific. They do not show that PPO or imitation learning is inherently unsuitable for orbital control. It is worth noting that larger datasets, alternative objectives, stronger baselines, or hybrid controllers could possibly change the result, subject to further research.

2.9 Data Preprocessing

Preprocessing was relatively limited. The scripts derived radius, decomposed velocity, recorded crossings, and evaluated later basin entry. I think this direct pipeline improves interpretability, although the absence of uncertainty modeling means small numerical or threshold changes were not systematically stress-tested.

Learning data required more preparation. Expert trajectories became state-action samples, and phase-conditioned variants also used explicit phase labels. It is worth noting that these pipelines were not standardized across methods, which may partly explain why the resulting counts cannot support a single comparative ranking.

2.10 Experimental Design

The experiments use controlled comparisons. Phase34 compares post-cross modes, Phase36B compares transfer families, and Phase37A compares six radial-timing variants. In my opinion, holding the benchmark fixed helps isolate design changes, although repeated use of the same grid possibly encourages benchmark-specific tuning.

Phase37B follows a narrower design. It tests four selected failures and eight regression successes under two settings. It is worth noting that this structure resembles checking a repair on both broken and working machines, because improvement matters only if known successes remain intact.

Representative figures explain mechanisms. Aggregate claims, however, come from stored counts and validated summaries rather than from individual trajectories. I think this distinction is important because a visually persuasive example can still be unrepresentative of the broader benchmark.

2.11 Result Validation

The critical artifacts passed regression checks. Environment files, smoke tests, workflow automation, and reproduction notes provide additional engineering support. It is worth noting that these checks strengthen traceability but do not replace an independent rerun of every experiment from a clean environment.

3. Results

3.1 Target-Radius Crossing Did Not Imply Recoverable Crossing

The baseline result is clear. The Phase31-style reference produced 8/24 target-radius crossings and 0/24 recoverable crossings. In my opinion, this is the strongest answer to the research question because the same controller looks partly successful under one metric and unsuccessful under the stricter one.

Metric choice changes interpretation. A radius-only score would count one third of cases, while recoverability counts none. It is worth noting that this contrast is supported by the available data, although it remains specific to the chosen thresholds and benchmark.

3.2 Post-Cross Synchronization Improved Recoverability

Phase34 changed the post-cross state. It preserved 8/24 crossings while increasing recoverable crossings from 0/24 to 8/24. I think this result suggests that downstream synchronization, rather than wider access to the target radius, produced the observed improvement.

No overspeed case was recorded. The eight already-crossing cases later entered the simulator-defined recoverability basin under radius_priority. It is worth noting that Figure 1 illustrates one case only, so the benchmark conclusion must rest on aggregate counts rather than visual appeal.

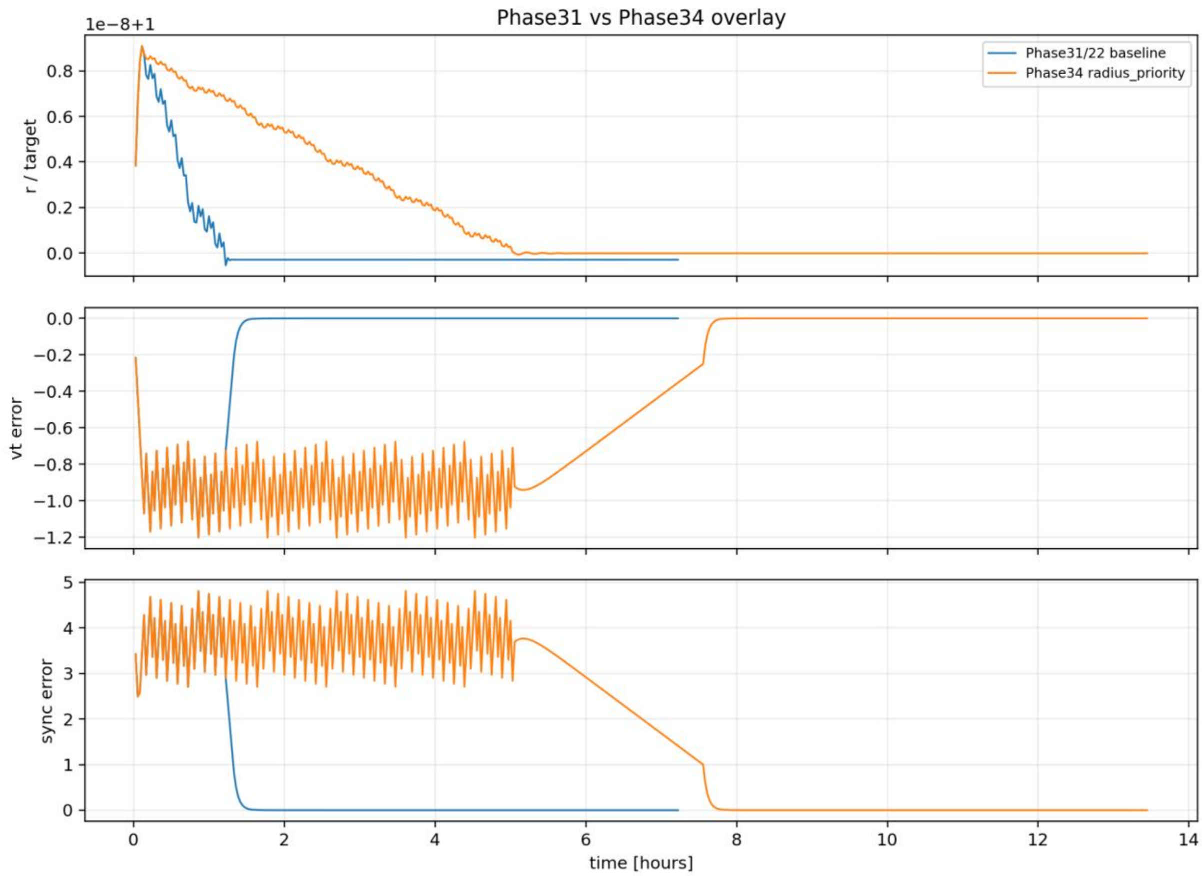

Figure 1. Representative post-cross trajectory for one benchmark case. The figure is illustrative rather than aggregate evidence. The case uses r0 = 1.0, initial_velocity_angle_deg = 175, and thrust_scale = 10000; benchmark-wide counts appear in Table 4.

3.3 Transfer-Family Variants Did Not Expand the Empirical Crossing Basin

Phase36B returned identical counts. Each of four families produced 8/24 crossings and 8/24 recoverable crossings, with no overspeed or instability flags. In my opinion, those results show preservation rather than expansion, because no additional tested initial condition entered the empirical crossing basin.

Equal counts can still hide differences. The trajectories may differ even when the final totals match across the selected grid. It is worth noting that the experiment does not establish equal robustness, equal control effort, or equal behavior beyond these cases.

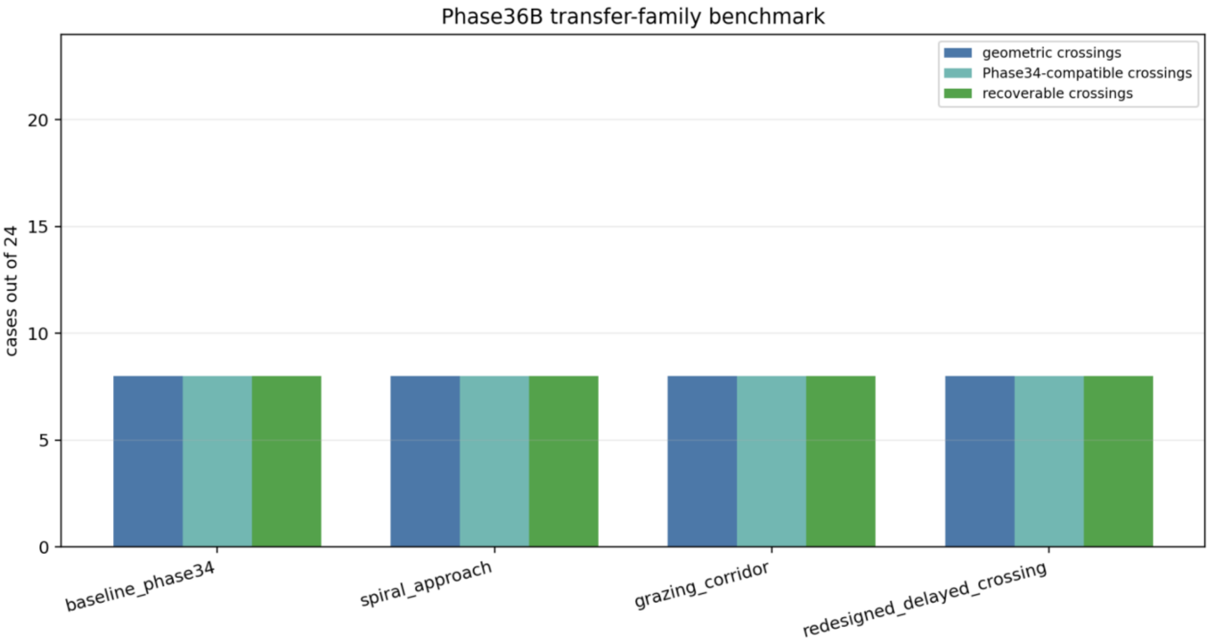

Figure 2. Phase36B transfer-family benchmark. Four families were evaluated on 24 cases each. Every family obtained 8/24 crossings and 8/24 recoverable crossings; none expanded the empirical crossing basin.

3.4 Radial-Timing Diagnostics

Phase37A tested six variants. Across 144 rollouts, none produced a new crossing among the 16 baseline non-crossing cases. I think this weakens the specific radial-timing hypothesis, although it does not prove that all possible radial schedules are ineffective.

Delayed variants preserved the known cases. In contrast, in my opinion, early and mid variants appear to have reduced the existing crossing set. It is worth noting that, like a classroom safety checklist, zero overspeed and instability flags only confirm compliance with coded rules, not physical safety or formal stability.

3.5 Weak-Tangential Subset Diagnostic

Phase37B used a smaller test. Weak tangential shaping produced 0/4 crossings in selected failures and preserved only 4/8 regression crossings. In my opinion, this resembles improving a student's practice score while causing previously mastered questions to be missed, which cannot yet count as net progress.

Closest approach improved in 3/4 cases. However, none of those selected cases crossed or became recoverable under the tested setting. It is worth noting that proximity, crossing, and recoverability measure different stages, so movement in one metric may not transfer to the next.

3.6 Learning-Based Baselines

The learning results were mostly negative. Behavior cloning, PPO fine-tuning, phase-conditioned imitation, soft phase conditioning, and minimal imitation produced zero crossings in their own evaluations. I think these failures possibly reflect compounding error, limited demonstrations, objective mismatch, or training choices, but the experiments do not isolate one cause.

One diagnostic reported five crossings. Learned phase stabilization still retained a simulator success label of false under its method-specific protocol. It is worth noting that these counts use different denominators and settings, so they cannot be compared directly with the unified 24-case benchmark.

Table 4. Main Results Summary

Evidence block	Scope	Target-radius crossings	Recoverable crossings	Safety / regression	Interpretation
Phase31-style reference	24 cases	8/24	0/24	Overspeed 0	Crossing did not imply recoverability
Phase34 radius_priority	24 cases	8/24	8/24	Overspeed 0	Existing crossing cases became recoverable
Phase36B transfer families	4 families x 24 cases	Each 8/24	Each 8/24	Overspeed 0; instability 0	No empirical basin expansion
Phase37A radial timing	6 variants x 24 cases	Best 8/24; 0/16 new	Best 8/24	Overspeed 0; instability 0	No new crossings under tested timing
Phase37B weak tangential subset	24 subset rollouts	Selected 0/4; regression 4/8	Selected 0/4; regression 4/8	Overspeed 0; instability 0	Selected failures remained unresolved
Learning-based methods	Separate method-specific evaluations	Mostly 0; one diagnostic reported 5	No learned-policy success shown	Success labels false	Diagnostic evidence only

4. Discussion

4.1 Why Geometric Crossing Was Insufficient

Radius captures only one coordinate. It does not guarantee low radial velocity, appropriate tangential velocity, or later basin entry. In my opinion, crossing the target radius is like touching a station platform from a moving train: contact occurs, but controlled arrival has not been achieved.

The Phase31-style result exposes this mismatch. Eight cases crossed, yet none later satisfied the recoverability conditions. It is worth noting that the conclusion is supported by the available data, but perhaps depends partly on the selected thresholds and deterministic environment.

4.2 Interpretation of the Phase34 Result

Phase34 solved a narrower problem. It did not increase the number of initial conditions that reached the target radius. I think its value lies in changing what happened afterward, allowing the eight existing crossing cases to enter the recoverability basin.

This separates two design tasks. Upstream control creates access, while downstream control regulates the state after access occurs. It is worth noting that Phase34 addresses only the second task for tested crossing-producing cases, leaving 16 non-crossing cases unresolved.

4.3 Why Crossing-Basin Expansion Remained Unresolved

The upstream challenge seems more difficult. It is worth noting that Phase36 preserved established results, whereas several Phase37 variants added no crossings and weakened prior successes. In my opinion, this pattern possibly means that simple timing shifts or weak tangential corrections are like turning one classroom dial at a time, while stronger coordinated changes remain untested.

Diagnostic improvement can be deceptive. Phase37B improved closest approach in several cases but still produced no selected-case crossing. It is worth noting that the available data cannot determine whether the limitation came from parameter range, controller structure, or the benchmark itself.

4.4 Role and Interpretation of AI Models

The AI methods remain relevant. PPO addresses continuous actions, while imitation learning uses demonstrations from the explicit controller [2], [3]. However, I think the cited methods do not guarantee success here, because long-horizon phase coordination can fail even when local predictions appear accurate.

Several explanations remain plausible. Limited demonstrations, compounding error, phase instability, reward design, model capacity, or evaluation mismatch could each contribute. It is worth noting that the present experiments do not separate these causes, making any categorical judgment about learning-based control premature.

Narrower AI tasks may work better. Recoverability prediction, failure classification, parameter search, or hybrid control possibly require less faithful long-horizon imitation. In my opinion, these ideas are promising hypotheses rather than completed results and remain subject to further research.

4.5 Scientific and Engineering Value

The main value is careful measurement. The framework separates geometric contact, later recoverability, and regression preservation across controller changes. I think this discipline helps prevent a good-looking trajectory or permissive metric from being mistaken for robust control performance.

Negative results can still teach us. In my opinion, several simple hypotheses failed under the tested settings, which helps narrow the next search. It is worth noting that these failures do not prove the ideas impossible; supported by available evidence, they only show where implementations, like lesson plans facing harder questions, stopped working.

4.6 Limitations and Critical Reflection

The simulator omits major physics. It excludes 3D motion, attitude coupling, perturbations, uncertainty, delays, and fuel depletion. In my opinion, these omissions are acceptable for mechanism isolation, but they sharply limit any claim about real spacecraft operations.

The recoverability thresholds are project-specific. They remain fixed across comparisons, yet their sensitivity was not tested. It is worth noting that alternative thresholds could perhaps change marginal classifications, even if the main crossing-versus-state distinction remains conceptually valid.

The benchmark is also small. Twenty-four designed cases do not represent a broad distribution, and no held-out grid or randomized robustness test is reported. I think the repeated 8/24 pattern may partly reflect benchmark structure, which makes broader generalization uncertain.

Learning comparisons remain uneven and difficult. In my opinion, different datasets, inputs, and protocols make any clean ranking potentially misleading for students and reviewers alike. It is worth noting that simulator safety flags, like passing a classroom checklist, do not establish physical safety, causal superiority, or deployment readiness, which remain subject to further research.

4.7 Future Work

The next step should protect successes. Candidate upstream changes should improve difficult cases while retaining the eight known crossing-producing cases. In my opinion, a regression-safe search over coast duration, radial push timing, magnitude, and tangential shaping offers a practical starting point.

Broader evaluation should follow. Randomized initial conditions, held-out cases, perturbations, threshold sensitivity, fuel use, and control effort would strengthen interpretation. It is worth noting that 3D dynamics and hybrid learned-explicit control should perhaps wait until the simpler crossing-generation bottleneck is better understood.

5. Conclusion

The research question received a cautious answer. Under this simplified benchmark, target-radius crossing was not sufficient for simulator-defined recoverable crossing. In my opinion, the 8/24 crossing and 0/24 recoverability result shows why geometric success can overstate actual state regulation.

Phase34 provided the clearest improvement. It preserved 8/24 crossings while increasing recoverable crossings to 8/24 through post-cross synchronization. It is worth noting that the improvement concerns downstream state quality, not expansion of the empirical crossing basin.

Other findings remain more tentative. Phase36 and Phase37 did not create new crossings without regression, while learning methods remained diagnostic. I think the study offers a useful evaluation framework, but its mission relevance, robustness, and generalization remain subject to further research.

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